

RESEARCH INTEREST

Networked Transport Systems; Optimal Control; Partial Differential Equations; Operator Learning; Smart Cities

EDUCATION

University of Southern California <i>Ph.D. Candidate, Civil Engineering; Advisor: Ketan Savla</i>	Los Angeles, USA <i>Sep. 2021 - present</i>
Southeast University <i>B.Eng., Transportation Engineering</i>	Nanjing, China <i>2016 - 2020</i>
University of California, Santa Barbara <i>Visiting Student; Host: Bassam Bamieh</i>	Santa Barbara, USA <i>Sep. 2025 - May 2026</i>

HONORS AND AWARDS

USC Department of Civil and Environmental Engineering Outstanding Research Assistant Award	<i>2025</i>
Professor Teh-Fu Yen and Dr. Shia-Ping Siao Yen Fellowship (\$20,000)	<i>2023</i>
USC Viterbi/Graduate School Fellowship (\$32,000)	<i>2021</i>

PUBLICATIONS

Refereed Journal Articles

8. **Li, Z.**, F. P. J. de Barros, & K. Savla (2025). Constrained bilinear optimal control of reactive evolution equations. *Under Review in Computer Methods in Applied Mechanics and Engineering*.
7. **Li, Z.**, A. S. Fokas, & K. Savla (2025). A complex spatial frequency approach to optimal control of linear evolution systems with general boundary conditions. *Conditionally accepted by IEEE Transactions on Automatic Control*.
6. **Li, Z.** & K. Savla (2024). Output-feedback adaptive model predictive control for ramp metering: a set-membership approach. *Under Review in Automatica*.
5. Han, K., J. Wen, Y. M. Nie, **Z. Li**, & S. Liu (2024). Exploring the sensing power of mixed vehicle fleets. *Transportation Research Part B: Methodological*, 190, 103066.
4. **Li, Z.**, M. W. Levin, X. Qu, & R. Stern (2023). A Network Traffic Model for the Control of Autonomous Vehicles Acting as Moving Bottlenecks. *IEEE Transactions on Intelligent Transportation Systems*, 24(9), 9004-9015.
3. Wang, S., **Z. Li**, & M. W. Levin (2022). Optimal policy for integrating Autonomous Vehicles into the auto market. *Transportation Research Part C: Emerging Technologies*, 143, 103821.
2. Kang, D., **Z. Li**, & M. W. Levin (2022). Evasion planning for autonomous intersection control based on an optimized conflict point control formulation. *Journal of Transportation Safety & Security*, 14(12), 2074-2110.
1. Wang, Y., **Z. Li**, Z. Ye, & Y. Zhang (2020). Nighttime safety evaluation for signalized intersections at the signal cycle level based on rear-end conflict models considering lighting and traffic conditions. *Traffic Injury Prevention*, 21(1), 87-92.

Refereed Conference Proceedings

6. **Li, Z.** & K. Savla. Adaptive model predictive control for traffic signal timing with unknown demand and parameters. In *the 2025 American Control Conference (ACC)*. July 2025 in Denver, Colorado.
5. **Li, Z.**, A. S. Fokas, & K. Savla. On linear quadratic regulator for the heat equation with general boundary conditions. In *the 63rd IEEE Conference on Decision and Control (CDC)*. December 2024 in Milan, Italy.
4. Pooladsanj, M., **Z. Li**, K. Savla, & P. Ioannou. Performance Evaluation of Ramp Metering under Different Traffic Measurement Scenarios. In *the 103rd Annual Meeting of Transportation Research Board Annual Meeting (TRB)*. January 2024 in Washington, D.C.
3. **Li, Z.** & K. Savla. Output-feedback Model Predictive Control for Ramp Metering. In *the IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*. September 2023 in Bilbao, Spain.

2. **Li, Z.**, M. W. Levin, R. Stern, & X. Qu. A Network Traffic Model with Controlled Autonomous Vehicles Acting as Moving Bottlenecks. In *the IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)* . September 2020.
1. **Li, Z.**, X. Zhu, X. Liu, & X. Qu. Model-based Predictive Variable Speed Limit Control on Multi-lane Freeways with a Line of Connected Automated Vehicles. In *the 22nd IEEE International Conference on Intelligent Transportation Systems (ITSC)*. October 2019 in Auckland, New Zealand.

Papers in Preparation

1. **Li, Z.**, Z. Zhou, & K. Savla. Extended Fourier neural operator for optimal control of evolution equations. In preparation.

TALKS AND POSTERS

3. “Predictive control for traffic networks with performance guarantees”, *Poster at Southern California Civil Engineering Symposium at USC, UC Irvine, and Pacific Southwest Region University Transportation Center Annual meeting at UC Berkeley*, 2025.
2. “A complex spatial frequency approach to optimal control of bounded spatially distributed systems”, *Talk at 44th Southern California Control Workshop*, 2024.
1. “Output-feedback ramp metering: model-based and model-free control”, *Poster at Southern California Civil Engineering Symposium at UCLA*, 2023.

GRANT WRITING

- Predictive Control for Transport Systems with Performance Guarantees (PI: Ketan Savla) 2025
Assisted in conceptualization, data preparation, original draft and editing.
Target program: National Science Foundation – Dynamics, Control and Systems Diagnostics.

TEACHING EXPERIENCES

- Teaching Assistant** at USC: CE 471 (Principles of Transportation Engineering) *Fall, 2023 - 2024*
20% students rated excellent.
- Teaching Assistant** at USC: CE 330 (Computer Methods in Civil Engineering) *Spring, 2024 - 2025*
33.3% students rated excellent. Course score (3.67/4).

ACADEMIC SERVICES

- Journal Reviewer:** *Automatica, Journal of Applied Mathematics and Physics (ZAMP)*.
- Conference Reviewer:** *IEEE Intelligent Transportation Systems Conference, IEEE Conference on Decision and Control, IFAC Modeling, Estimation and Control Conference, TRB Annual Meeting*.

MENTORING AND LEADERSHIP

- Research coach at USC Viterbi Summer Institute *Summer 2022 - 2023*
Mentored Native American, Black, and Latinx engineering incoming undergraduate students at USC.
- Supervisor for the Center for Undergraduate Research in Viterbi Engineering *2022 - 2024*
Mentored undergraduate research assistants Emma Silverstein, Kathryn Ruiz, and Aryan Shukla.
- Graduate Student Research Advisor *2024 - present*
Mentored Master students Yajun Liu (pursuing PhD at ASU), Zhengbang Liu (Joined BYD), Xinyi Wu (applying for PhD), and Zixiong Wang (Joined HNTB).

ACTIVITIES

- Active member of UMN Running Club (2019 – 2020) and SC Triathlon Club (2021 – 2023)
- College Athlete of the Orienteering Team at Southeast University (2016 – 2019)